

μPD7202xx

Board Layout Guidelines

Introduction

The μPD720201 and μPD720202 are USB 3.0 host controllers from Renesas Electronics, while the μPD720210 and μPD720211 serve as USB 3.0 hub controllers.

This document provides best practices for designing printed circuit boards (PCBs) using USB 3.0 host and hub controllers. It explains critical layout considerations such as trace length limitations, impedance control, ground plane requirements, and routing strategies to maintain signal integrity for high-speed data transmission. Additionally, it includes detailed recommendations for component placement, crosstalk mitigation, and power/ground design to ensure reliable performance in USB 3.0 systems.

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1. USB 3.0 Board Design Guidelines

Renesas USB 3.0 devices are engineered for ease of integration into USB systems; however, board design requires careful consideration of high-speed data transfer and high-frequency circuitry. Failure to address these factors can result in system-level issues.

These guidelines serve as a reference for designers implementing USB 3.0 devices on motherboards, mainboards, PCs, consumer electronics auxiliary boards, and add-in cards.

Note: Recommendations in this document assume the use of FR-4 as the PCB material. While other materials suitable for high-speed transmission may allow more relaxed conditions, they are not covered in the document.

Diagrams in this guide may include “O” or “X” symbols, where “O” indicates recommended practices and “X” denotes practices to avoid.

1.1 Trace Length Limitations for SuperSpeed and PCIe Signals

USB 3.x devices impose specific trace length constraints on SuperSpeed (SS) and PCIe signal lines due to their high-speed data transmission requirements. USB 3.0 hubs limit both downstream and upstream ports, while peripheral controllers restrict upstream ports. Host controllers apply limitations to downstream SS signal lines and PCI Express 2.0 connections to the system.

For FR-4 PCBs, the recommended trace length for SS and PCIe signals is 10cm (4in) or less. Actual maximum lengths vary by layout but are typically estimated at:

- USB 3.x SuperSpeed Downstream ports: up to 28cm (11in)
- USB 3.x SuperSpeed Upstream ports: up to 12.7cm (5in)
- PCIe 2.0 signals: 15.2cm (6in) on motherboards and approximately 10cm (4in) on daughter cards.

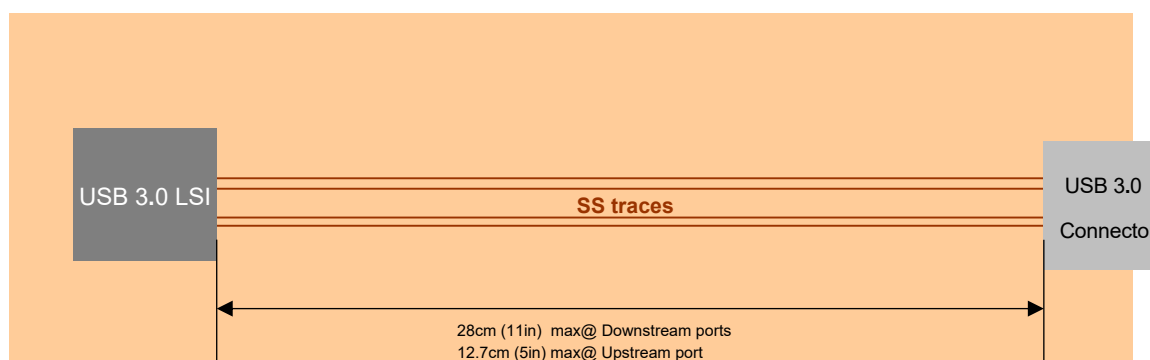


Figure 1. Estimated Maximum Trace Length for SuperSpeed Signals

To maintain signal integrity for USB SS traces ranging from 10cm (4in) to 28cm (11in), strict adherence to the design rules in this document is essential.

USB traces terminated with external connectors are more susceptible to signal degradation, such as crosstalk, compared to PCI Express traces. For extended SS trace lengths, consider replacing PCB traces with low-loss cable wiring.

Note: On boards functioning as USB hosts with PCI Express data transfer, PCIe traces may exceed SS trace lengths because USB traces typically experience greater degradation than on-board PCIe signals.

If SS trace lengths exceed 10cm (4in), verify feasibility through the following steps:

1. Create a test channel on the PCB and confirm a differential impedance of $90\Omega \pm 10\%$ using the TDR method.
2. Measure differential signal attenuation (SDD21) up to 7.5GHz with a network analyzer.
3. Connect the test channel to a USB 3.0 add-in card or equivalent device and perform SS compliance tests as defined by USB Implementers Forum (USB-IF).
4. Use test results to evaluate PCB trace performance.

1.2 Installing AC Coupling Capacitors for Signal Lines

AC coupling capacitors must be placed only on SuperSpeed TX signal traces, not on RX traces. Use ceramic chip capacitors with a capacitance of 0.1μF (100,000pF) per trace.

Because chip capacitors inherently exhibit inductance, which can create high impedance at high frequencies and lead to USB data transfer failures, select capacitors with an inductance of 0.2nH or less to ensure optimal SuperSpeed performance.

Place AC coupling capacitors symmetrically near the receptacle, as shown in the Figure 2.

Note: This recommendation also applies to PCI Express signal lines.

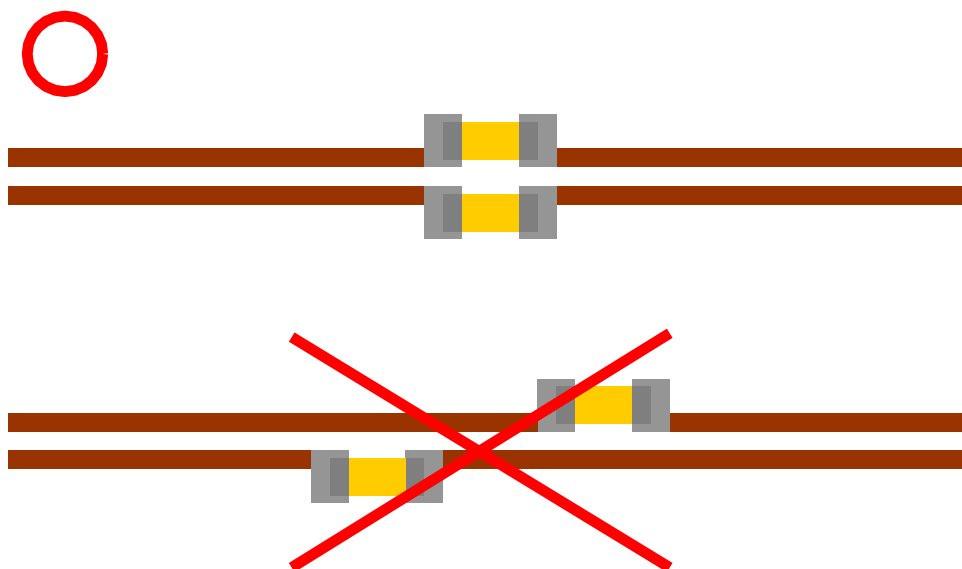


Figure 2. Placement of AC Coupling Capacitors for TX Signals

Note: For each differential pair, use capacitors from the same production lot. Mixing different lots can introduce common-mode noise and degrade signal integrity.

1.3 Crossing Traces

When a signal trace crosses another signal trace or a power supply plane, ensure at least one ground plane separates them. Crossing without a ground plane layer can introduce crosstalk, significantly degrading system performance.

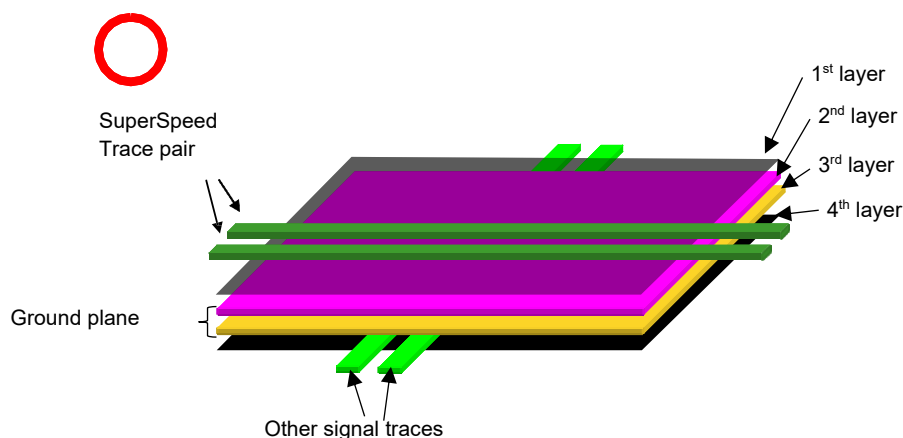


Figure 3. Recommended Signal Trace Crossing

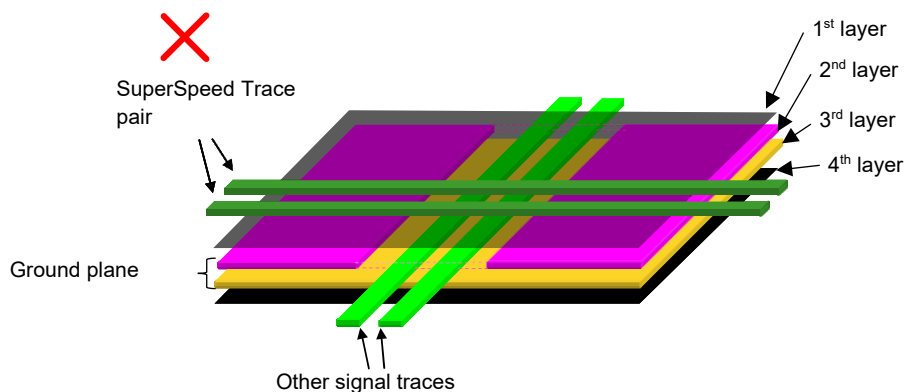


Figure 4. Depreciated Signal Trace Crossing

In Figure 3, the SuperSpeed differential pair is routed on the first layer, while the intersecting signal is placed on the fourth layer. This configuration ensures a stable ground plane between the signals, effectively mitigating crosstalk. Conversely, in Figure 4, the intersecting signal resides on the second layer, leaving no ground isolation and making crosstalk unavoidable. For more information, see “Routing Guidelines for USB Differential Signal Pairs”.

1.4 Limiting Adjacent Traces

Crosstalk between adjacent traces can also degrade system performance. To minimize this, maintain an adequate gap between traces. The minimum spacing should be $\geq 0.5\text{mm}$ (20mils) or four times the dielectric height, whichever is greater, as shown in Figure 5.

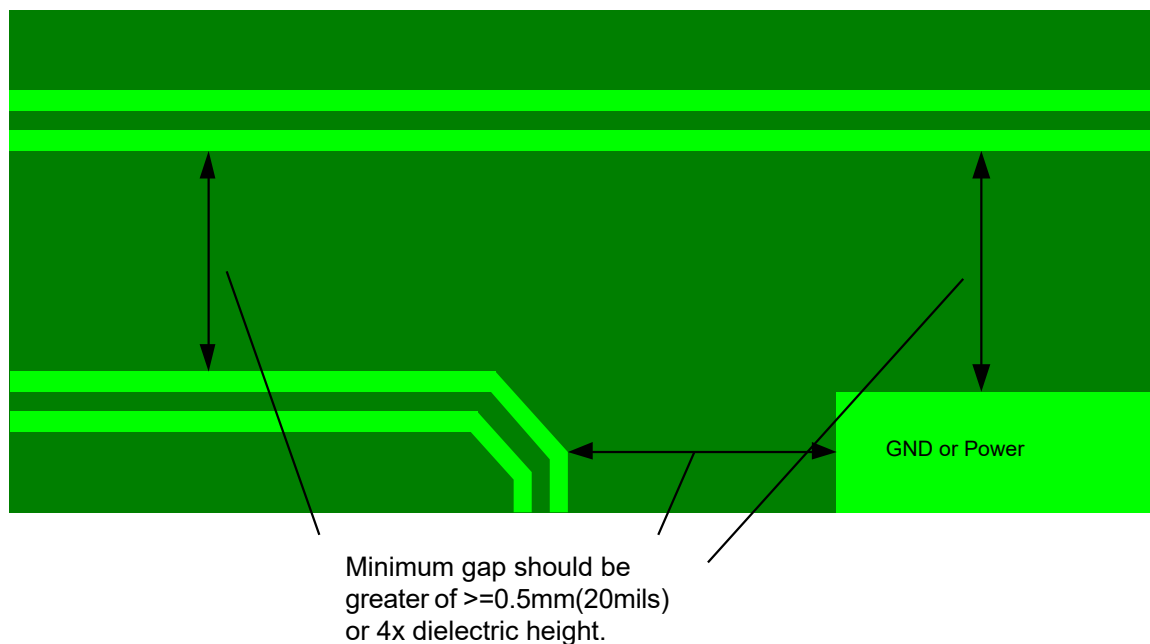


Figure 5. Distance Limitation with Adjacent Signals

Avoid routing long signal traces side by side with different signals because this can lead to excessive crosstalk. For more information, see “Routing Guidelines for USB Differential Signal Pairs”.

1.5 PCB Substrate

A four-layer PCB or higher is required for implementing USB 3.0 devices. Recommended layer assignments for a four-layer board are shown in Table 1.

Table 1. Recommended PCB Layer Signal Assignment

Layer	Example 1	Example 2
First layer	Data signal, clock	Power supply, control signal
Second layer	GND	Power supply, GND
Third layer	Power supply, GND	GND
Fourth layer	Power supply, control signal	Data signal, clock

All data signal traces must be fully isolated by adjacent ground planes to ensure signal integrity.

1.6 Routing Guidelines for USB Differential Signal Pairs

This section outlines best practices for routing USB SuperSpeed differential pairs. Due to PCB size constraints, achieving all layout goals may be challenging; therefore, prioritize the following items in order of importance as shown in Table 2.

Table 2. Priorities in Layout Design

Priority	Item	Item Number
1 (Highest Priority)	Ground Plane	1.6.1
2	Line Impedance	1.6.2
3	Equal trace length for differential pairs	1.6.3
4	Symmetric Routing	1.6.3、1.6.4
5	Avoid Changing Signal Layer	1.6.5
6	Eliminate signal stubs	1.6.6

Aim to satisfy as many high-priority items as possible to maintain optimal signal quality.

1.6.1 Ground Plane

To ensure optimal signal integrity, large continuous ground areas are required on layers adjacent to differential signal traces. All signals should be routed entirely over a ground plane. The ground layer must remain free of signal lines, voids, and power patterns that could compromise differential signal performance.

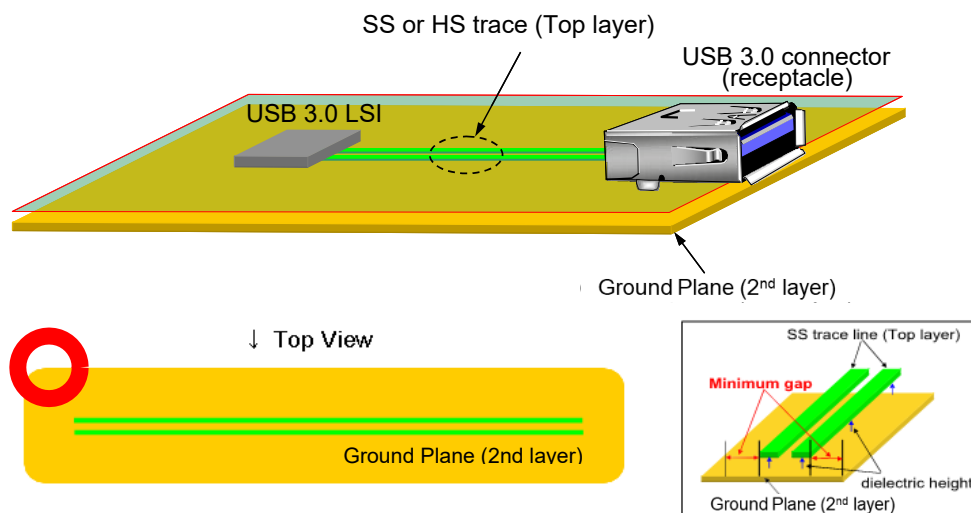


Figure 6. Differential Signals and Ground Plane Relationships

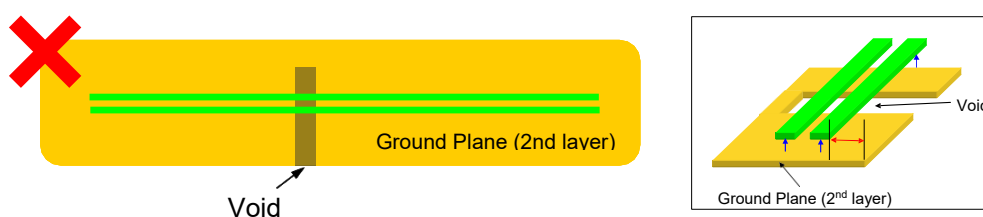


Figure 7. Ground Plane with Voids (Example 1)

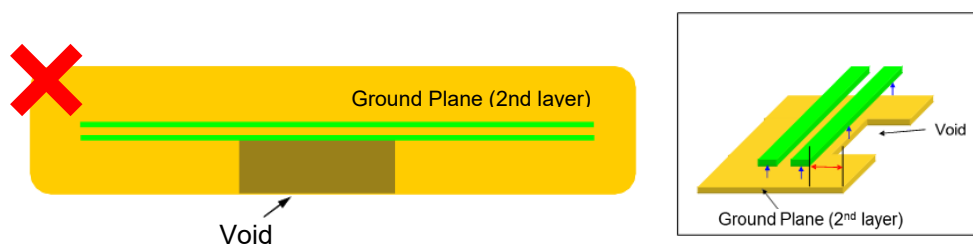


Figure 8. Ground Plane with Voids (Example 2)

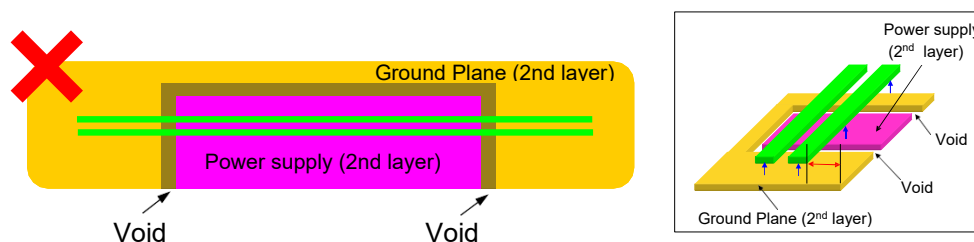


Figure 9. Ground Plane with Power Supply Pattern

1.6.2 Line Impedance

USB differential pairs must maintain an impedance of $90\Omega \pm 10\%$. To prevent mismatches, keep trace width and pair spacing uniform, and ensure adequate clearance from adjacent traces as specified in “Limiting Adjacent Traces.”

For PCI Express:

- PCIe 1.1 – 100Ω nominal
- PCIe 2.0 – 85Ω nominal (90Ω up to 120Ω allowed if design constraints apply)

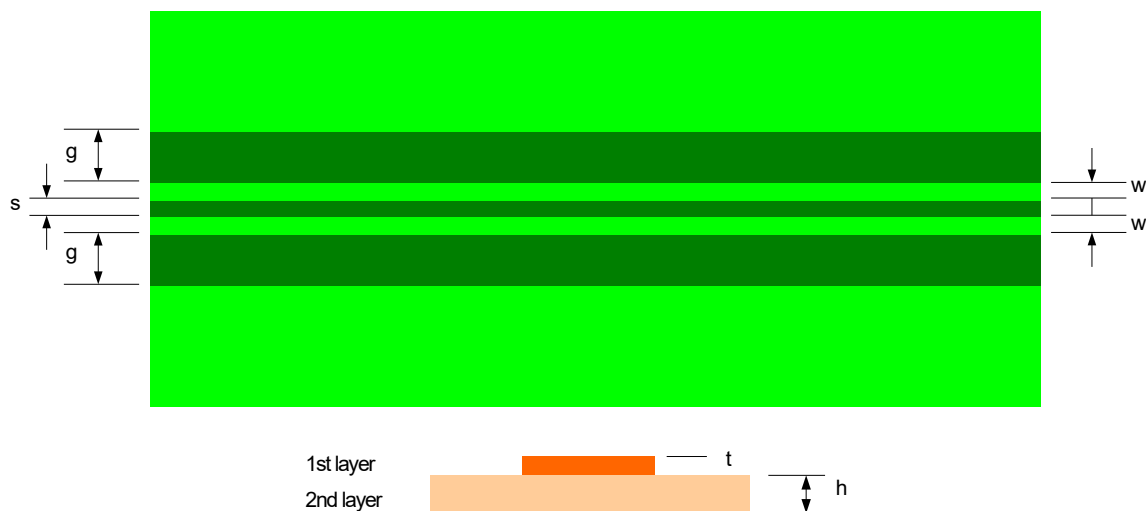


Figure 10. Dimensions of Differential Signals

Table 3. Recommended Dimensions for 90Ω Differential Impedance

ϵ_r	Dielectric constant	4.1 (FR-4)
h	Layer height of the second and first floors	0.1mm (4mil)
t	Metal thickness	$38\mu\text{m}$ (1.5mil)
w	Trace width	0.15mm (6mil)
s	Pair spacing	0.10mm (4mil)
g	Clearance	0.50mm (20mil)

1. FR-4 dielectric constant varies by vendor and board location.
2. Differential impedance is not simply twice single-ended impedance; verify using TDR.
3. To minimize EMI, pair spacing should not exceed trace width.
4. Avoid nickel-plated metal on outer layers due to low conductivity.

1.6.3 Differential Signal Routing

Route differential pairs on the same layer with consistent length, width, and spacing to maintain stable differential impedance. Match trace lengths within 0.12mm (5 mil). Symmetrical routing is recommended; if asymmetry occurs, adjust lengths near the connector.

Symmetry between different pairs (for example, SSTX and SSRX) is not required.



Figure 11. Differential Signal Routing (Example 1)

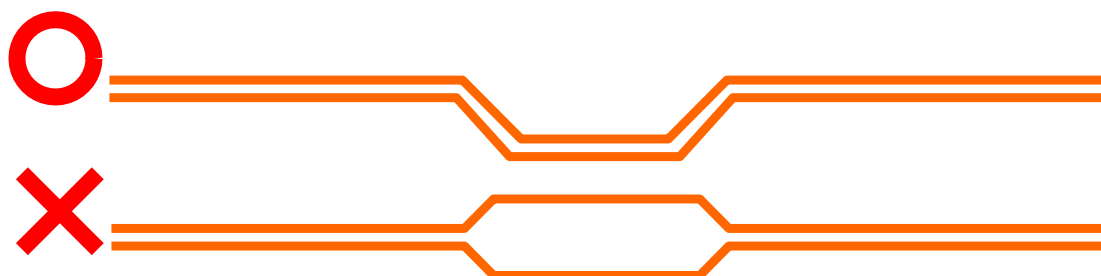


Figure 12. Differential Signal Routing (Example 2)

In Figure 12, the bottom trace is more geometrically symmetrical but the top trace better preserves uniform spacing and impedance along the path.

1.6.4 Trace Bending

Minimize trace bends – preferably no more than two – except near the USB connector or USB LSI. Avoid sharp angles like in Figure 13 that can introduce noise; use “45° or rounded bends” as shown in Figure 14.

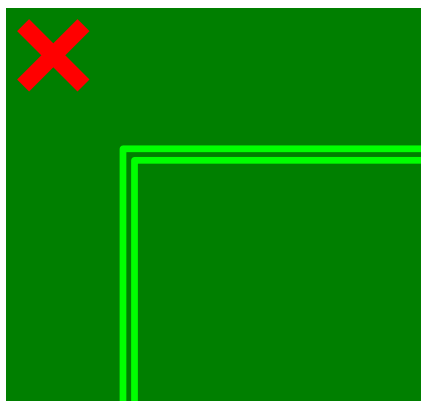


Figure 13. Trace Bending That is Not Recommended

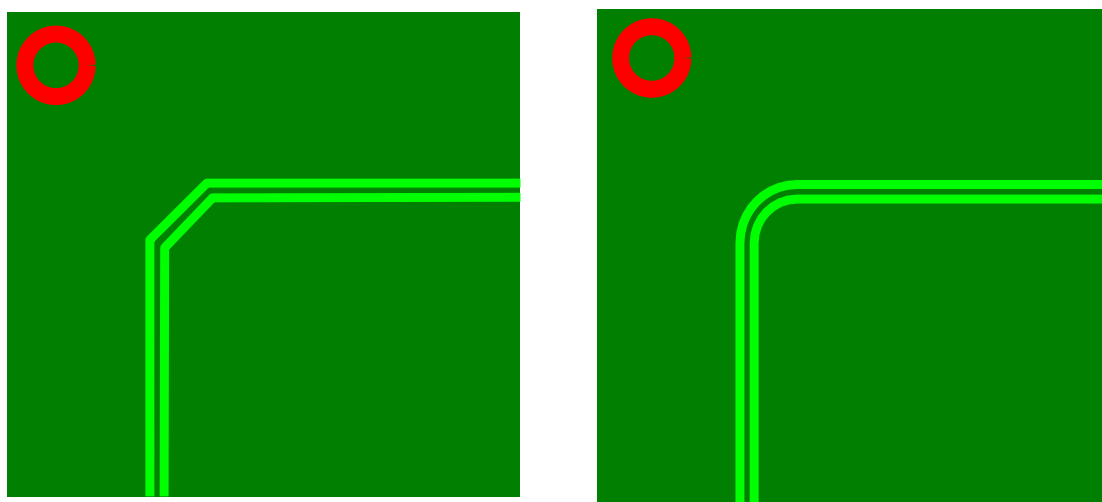


Figure 14. Recommended Trace Bending

Maintaining consistent spacing between differential pairs takes priority over minor length mismatches caused by bends. Adjust lengths as needed to meet the tolerance defined in “Differential Signal Routing”.

1.6.5 Changing Signal Layer

When a signal trace must transition to another layer, insert sufficient vias to maintain ground continuity (see Figure 15).

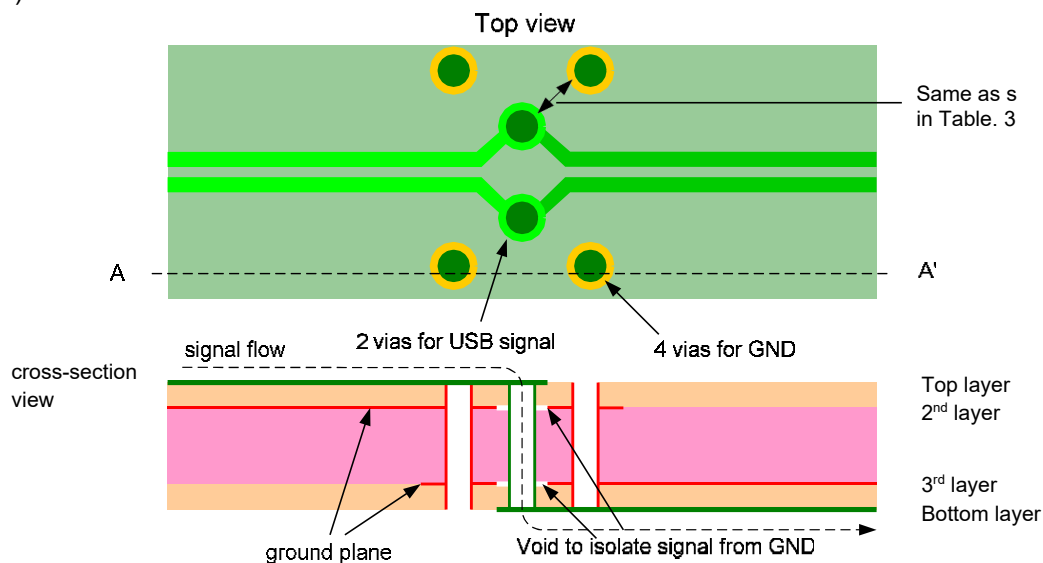


Figure 15. How to Change the Wiring Layer of USB Differential

Via diameter in Figure 15 should approximately match the trace width (w). Remove unnecessary metal patterns and create inner-layer voids to minimize impedance discontinuities.

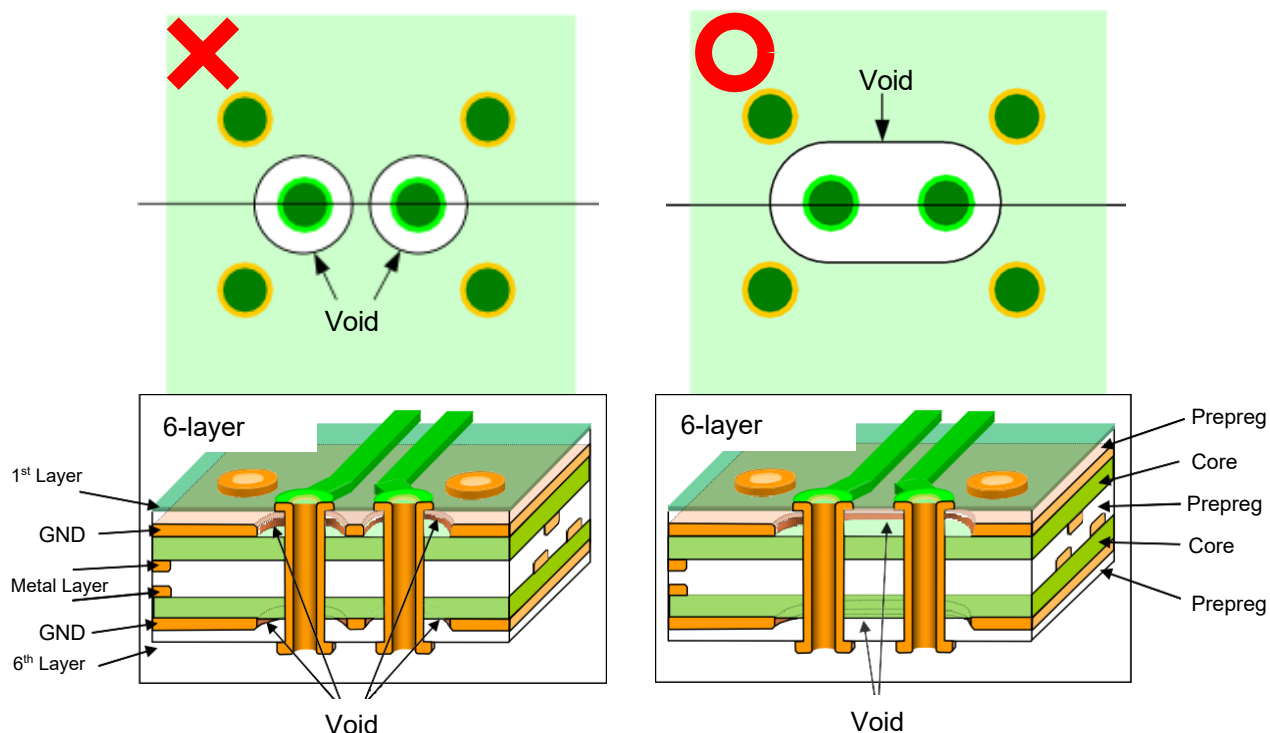


Figure 16. Layout of the Inner Layer when Changing the Wiring Layer of the USB Differential

As shown in Figure 16, improper void placement can cause impedance variation. Use the recommended void layout shown on the right to ensure signal integrity.

1.6.6 Installing Signal Stubs

Signal stubs should be avoided. Do not introduce unintended stubs on any trace.

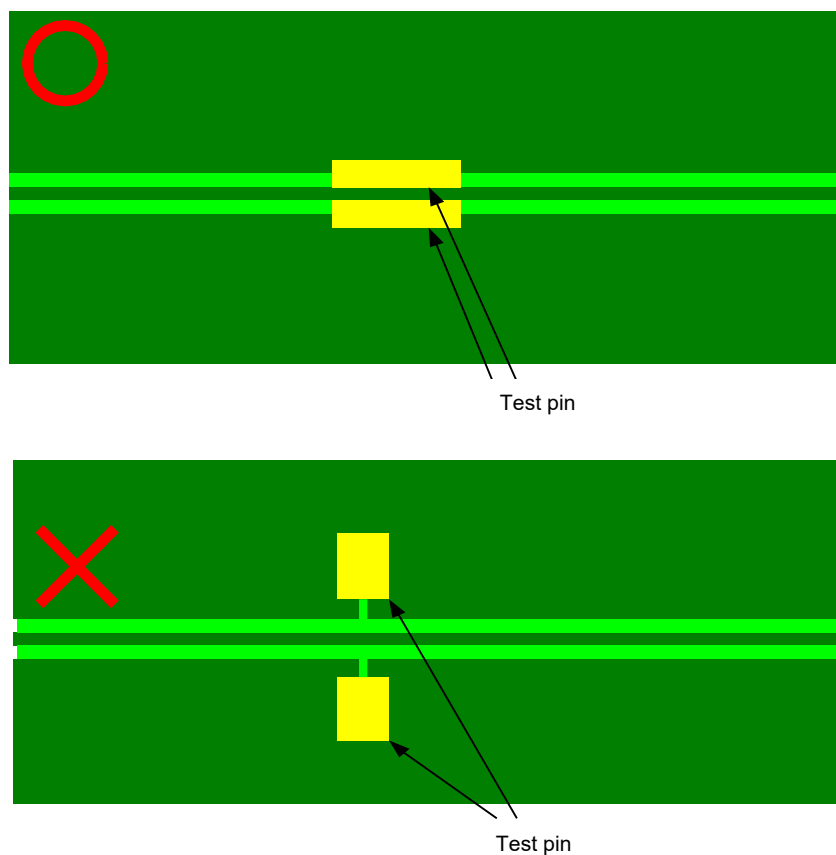


Figure 17. Signal Stub Placement

When adding test pads, align them directly over the signal trace to prevent stub formation (see Figure 17).

2. Routing Around the USB Receptacle

When connecting to a USB receptacle, maintain signal trace impedance, avoid stubs, and eliminate routing that may cause EMC issues. Crosstalk must also be considered.

2.1 Crosstalk Between USB Signals

In USB 3.0, near-end crosstalk typically occurs in three cases:

- SSTX → HS in RX mode
- SSTX → SSRX
- HS → SSRX in TX mode

To minimize interference, avoid placing SSTX, SSRX, and HS traces in close proximity. This is especially critical when connecting to a USB 3.0 hub because all communication modes (SSTX, SSRX, HS) may be active simultaneously.

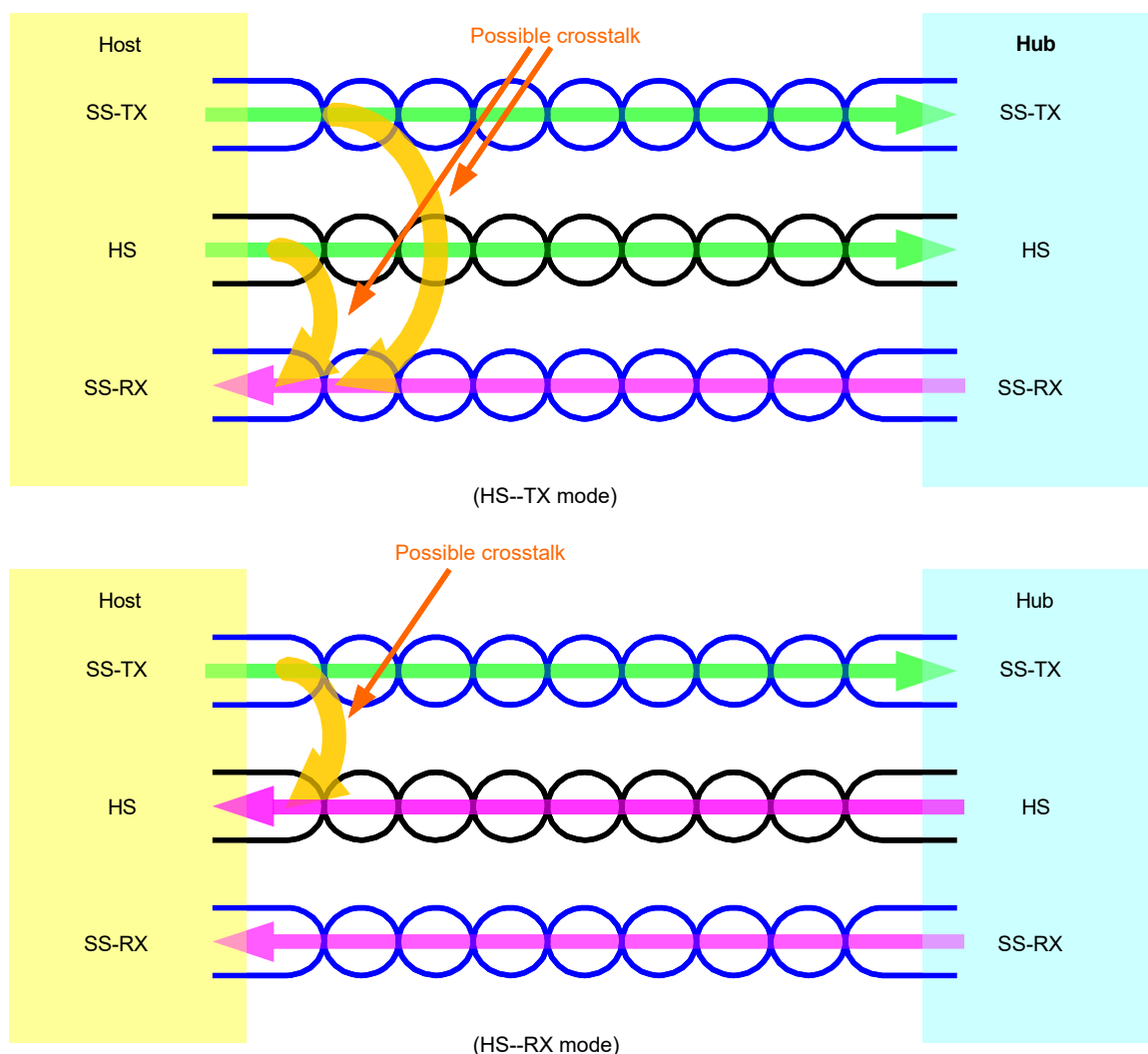


Figure 18. Crosstalk Possibilities

2.2 Signal Routing Around the USB Receptacle

This section provides examples of trace layouts near USB receptacles. For cable shielding, use AC isolation to the PCB ground – typically by selecting an appropriate inductor – rather than connecting the cable shield directly to the ground plane.

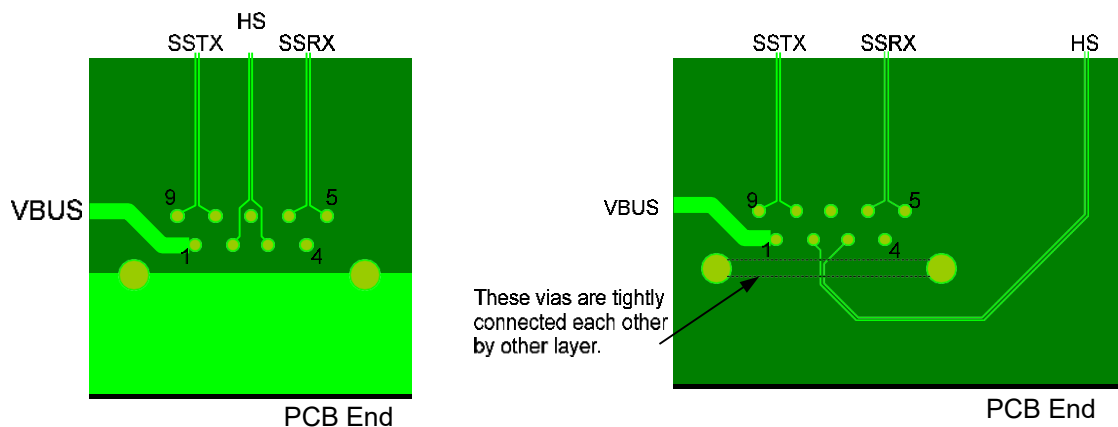


Figure 19. Standard-A Receptacle Connection Example

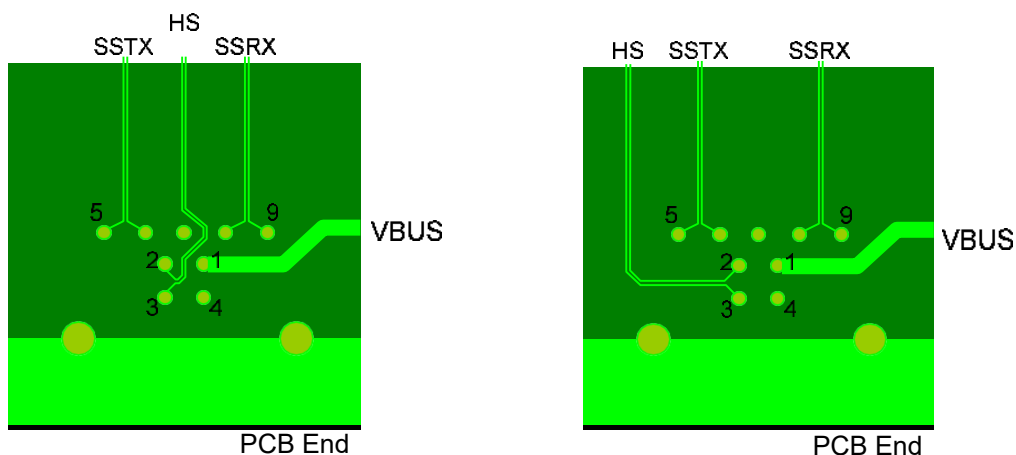


Figure 20. Standard-B Receptacle Connection Example

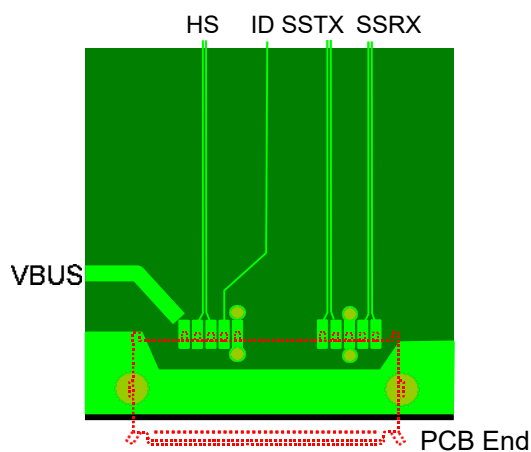


Figure 21. Micro-B Receptacle Connection Example

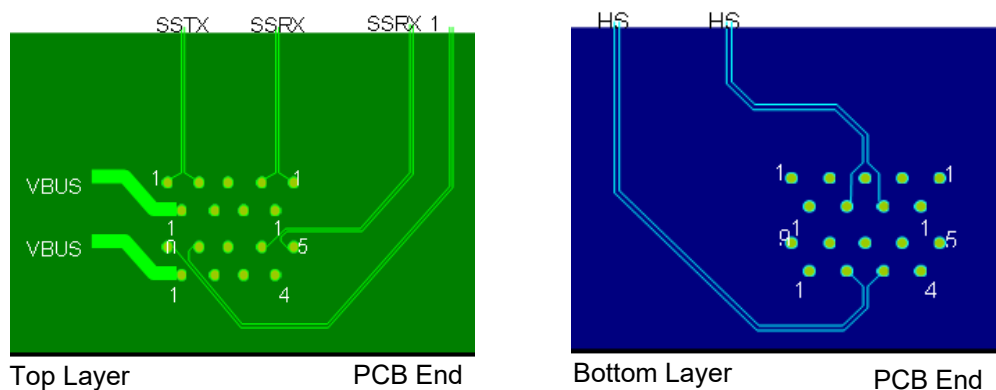


Figure 22. Standard-A Double-Stack Receptacle Connection Example

When using a Standard-A double-stack receptacle, consider potential crosstalk caused by its mechanical structure. For receptacles other than surface-mount types, incorporate voids to minimize crosstalk within the PCB.

2.2.1 Countermeasures Against Crosstalk in the USB Receptacle Area

When using two ground planes around a USB 3.0 receptacle – one for the earth pattern and the other for the cable shield at the same DC level – maintaining the differential impedance of the SuperSpeed (SS) traces is critical. To achieve this, remove metal patterns around the differential pair pins and surround them with voids, as shown in Figure 23 and Figure 24. Additionally, do not route any traces between the differential pair pins to preserve impedance.

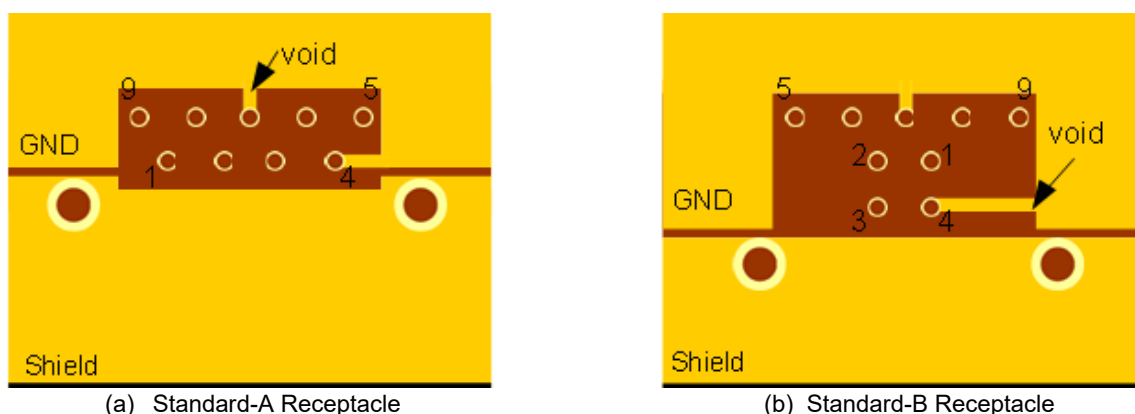


Figure 23. Void Entire Area Around Pins

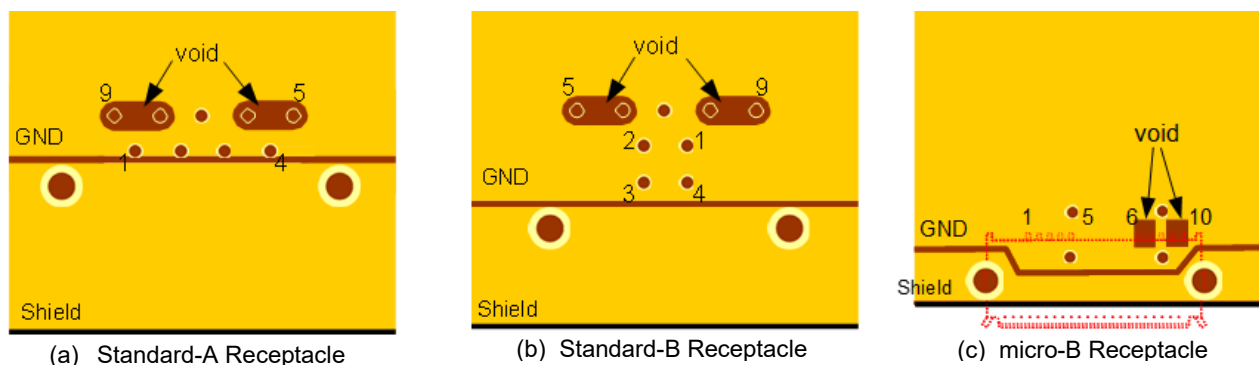


Figure 24. Void Only Around SS Pins

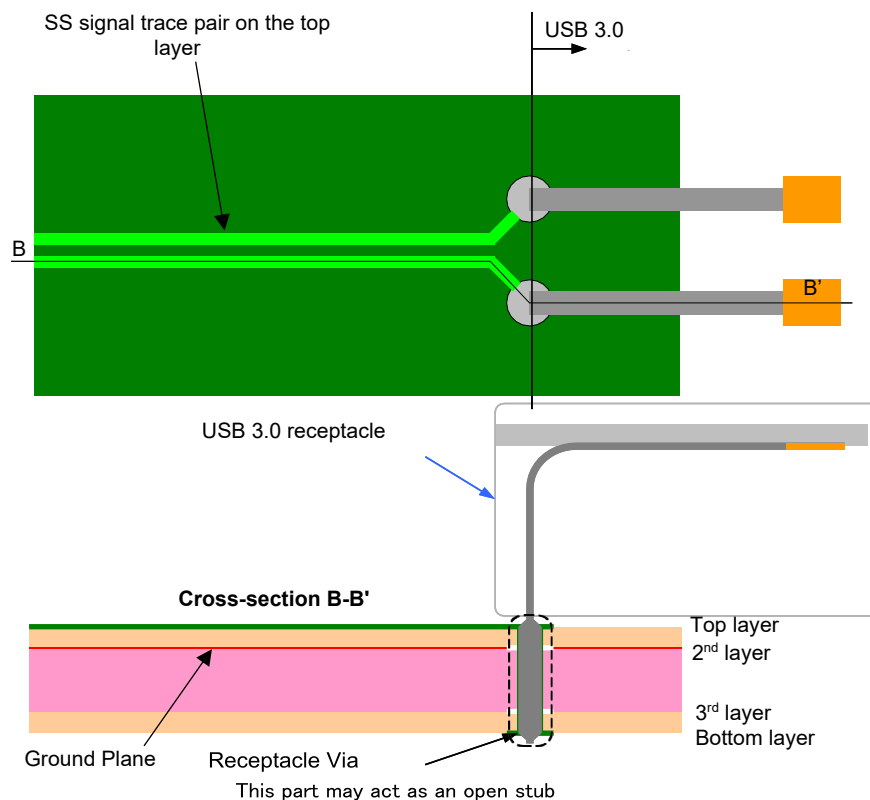


Figure 25. Example of SS Trace Connection to USB 3.0 Receptacle (Case 1)

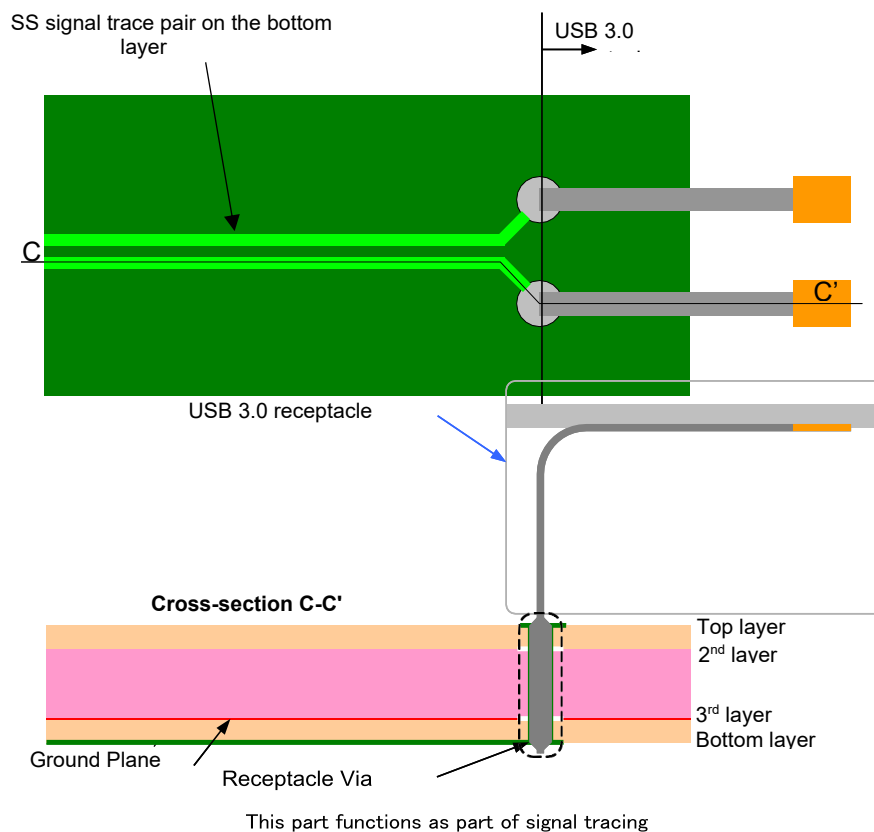


Figure 26. Example of SS Trace Connection to USB 3.0 Receptacle (Case 2)

For Figure 26, if the USB 3.0 device is located on the top layer, routing layer changes will be required.

3. Signal Routing Around USB Device

Similar to “Routing Around the USB Receptacle,” crosstalk must be carefully managed when routing signals around a USB device. Avoid narrow traces that can introduce parasitic inductance across power and ground planes. Ensure adequate current delivery to each power pin because voids around ground vias beneath the device may impact performance.

For SuperSpeed data transfers, do not split analog and digital ground planes because this can cause ground bounce and potentially damage the device. Instead, use a single and continuous ground plane. If multiple ground layers exist, provide sufficient vias to maintain ground stability.

Power planes should be separated by voltage level, with generous areas allocated for each. For noise suppression, insert ferrite beads between analog and digital power supplies of the same voltage. Keep analog power patterns isolated from digital traces, including signal lines, to prevent interference with analog circuitry.

Place decoupling capacitors as close as possible to the USB device’s power domain and ensure a solid ground connection.

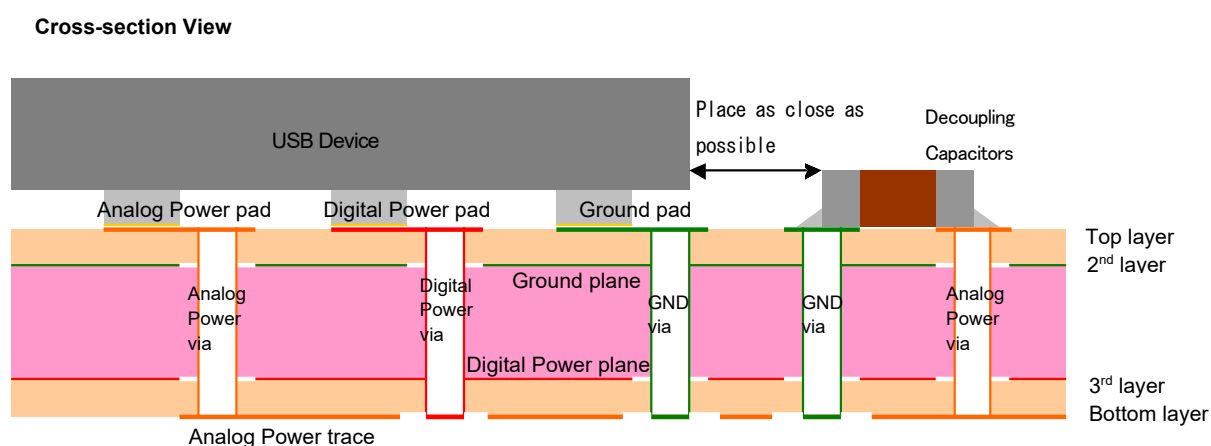


Figure 27. Example of Decoupling Capacitor Placement for Analog Power Supplies

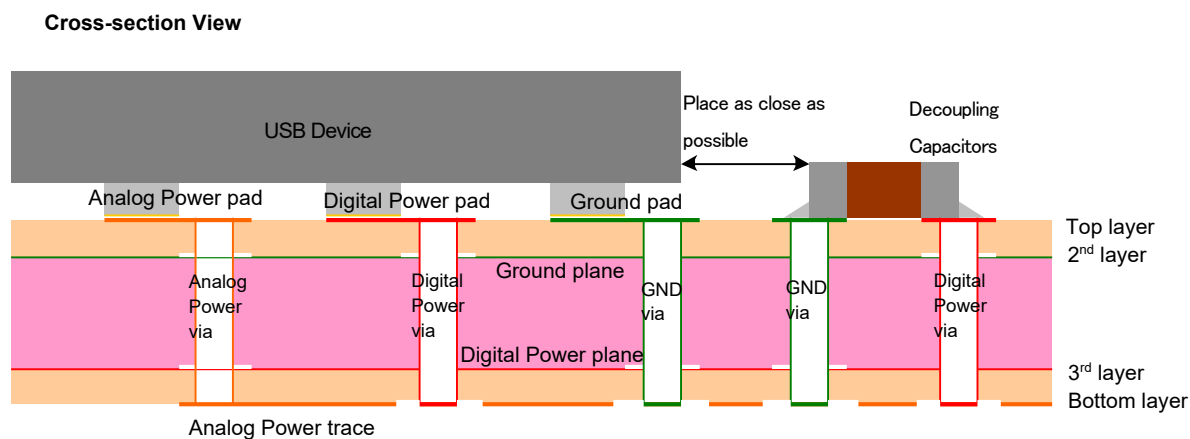


Figure 28. Example of Decoupling Capacitor Placement for Digital Power Supplies

The symmetry of the differential pair is inevitably disrupted at the USB device terminals, causing trace length variations. To meet the length tolerance specified in the “Differential Signal Routing”, adjust the trace length near the connector as illustrated in Figure 29.

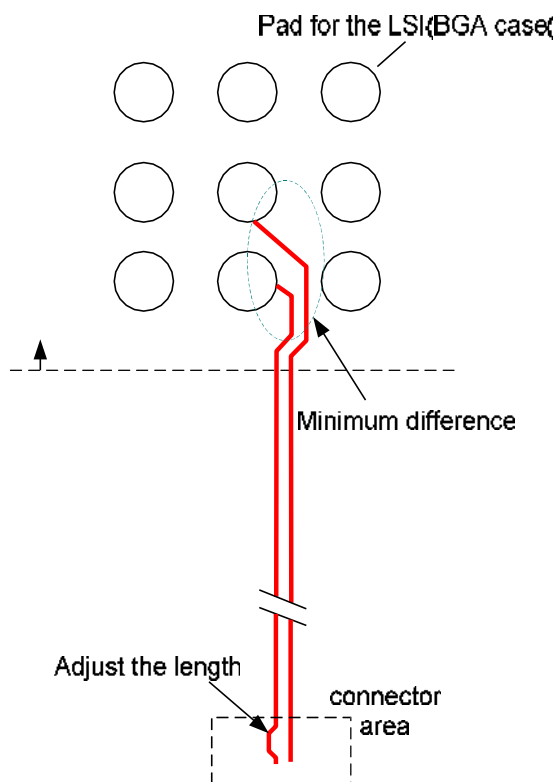


Figure 29. Example of Differential Signal Trace

3.1 Placement of RREF Resistors

For optimal USB system performance, connect the RREF pin to ground through a high-precision external resistor (tolerance less than 1%). Minimize the DC current path length and ensure isolation from digital noise. The RREF resistor should be placed on the same layer as the USB device for best results.

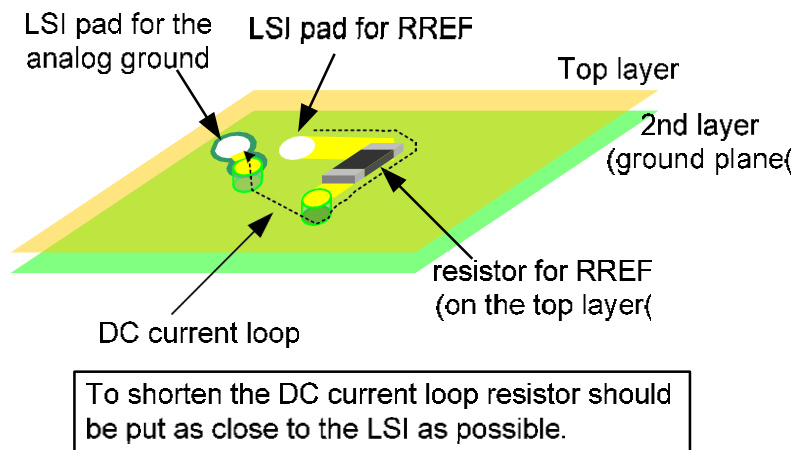


Figure 30. RREF Resistor Placement

4. USB 3.0 Board Design FAQs

This section provides a summary of frequently asked questions about USB 3.0 board design.

4.1 High-Speed Data Transmission Lines

Table 4. Key Questions and Answers for High-Speed Data Lines

	Question	Answer
1	What should be considered when changing the routing layer of a differential pair?	Ensure a ground plane is adjacent to the trace after the layer change, and place at least one ground via per trace near the transition to maintain impedance continuity.
2	What is the allowable length difference between traces in a differential pair?	0.12mm(5mil)
3	If the trace length from the device to the connector exceeds 8 inches, how can performance be maintained?	Use a material with lower attenuation than FR-4, such as a polyimide flat flexible cable. If using such a material, maintain proper impedance and dimensions for ground and VBUS.
4	What is recommended when a differential pair intersects another trace?	Insert a ground plane between the differential pair and the other trace.
5	Can the power plane serve as a reference plane?	No. Even with AC bypass capacitors, the power plane cannot act as a reference for USB 3.0 signals. Use a ground plane instead.
6	Should decoupling capacitors maintain symmetry in differential pairs?	Yes, symmetry is recommended to minimize common-mode noise.
7	What are key considerations for routing differential pairs on inner layers?	Account for different dimensions compared to surface layers and implement crosstalk protection.
8	Why is there a length limit for high-speed signal traces?	FR-4 is unsuitable for high-frequency circuits. At Gbps speeds, signal attenuation becomes significant, reducing system performance. For this reason, a trace length limit is set.

4.2 Power Supply/Ground

Table 5. Key Questions and Answers for Power/Ground

	Question	Answer
1	What is the appropriate width for a power supply trace?	It depends on PCB structure. The width must support acceptable I-R drop and low AC impedance to ground—the thicker, the better.
2	Should power or ground be prioritized?	Ground. Poor grounding degrades system performance.
3	Is it necessary to split analog and digital grounds?	For PCBs with high-speed data (>3 Gbps), splitting grounds generally introduces more disadvantages than benefits.

4.3 RREF

Table 6. Key Questions and Answers for RREF

	Question	Answer
1	What are best practices for placing RREF resistor?	Keep the path(including the RREF terminal, the AVSS terminal of the USB device, and the external resistor) short and completely isolated from power traces, planes, and signal lines. Place the external resistor on the same layer as the USB device and surround it with ground to reduce low-frequency noise.

4.4 Other

Table 7. Key Questions and Answers

	Question	Answer
1	How many layers should a PCB have?	More than four.
2	Can materials other than FR-4 be used?	Yes, but at your own risk.

5. Revision History

Revision	Date	Description
1.00	Mar 20, 2026	Initial release

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